

Information Retrieval Assignment Report-

HW6

1. Overview

This assignment combines scores from six retrieval models as features to train a Linear Regression model that predicts document relevance. The trained model then ranks documents by predicted relevance score.

The Cranfield dataset (1,400 documents, 225 queries) was used for all experiments.

2. Feature Matrix

Six retrieval models were used as features:

Feature	Model	Description
x1	Okapi TF	Term frequency with length normalization
x2	TF-IDF	Term frequency with inverse document frequency
x3	BM25	Best Matching 25 ($k_1=1.2$, $b=0.75$)
x4	Laplace LM	Language model with add-1 smoothing
x5	Jelinek-Mercer LM	Language model with interpolation ($\lambda=0.7$)
x6	ES Built-in	Elasticsearch built-in BM25

Feature Matrix Statistics:

Metric	Value
Total rows	22,500
Queries	225
Documents per query	100
Positive examples (relevant)	114 (0.5%)
Negative examples (irrelevant)	22,386 (99.5%)

3. Learned Coefficients (5-fold Cross Validation)

Average Coefficients across 5 folds:

Feature	Coefficient	Interpretation
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TF-IDF	+0.000659	Most important positive weight
BM25	+0.000646	Second most important
Okapi TF	+0.000264	Moderate positive weight
Jelinek-Mercer	+0.000005	Neutral (near zero)
Laplace	-0.000004	Neutral (near zero)
ES Built-in	-0.000738	Negative weight (degrades prediction)

Analysis of Coefficients:

- TF-IDF and BM25 had the highest positive coefficients, consistent with their strong individual performance
- ES Built-in had a negative coefficient, meaning higher ES scores actually lowered the predicted relevance
- Laplace and Jelinek-Mercer coefficients were near zero, indicating they contributed little to the prediction

4. Results (MAP Scores)

Rank	Model	MAP Score
1	TF-IDF	0.008606
2	Okapi TF	0.008157
3	BM25	0.007992
4	ES Built-in	0.007986
5	Jelinek-Mercer	0.006262
6	ML Linear Regression	0.006186
7	Laplace	0.005515

5. Analysis

5.1 Why ML Did Not Outperform Individual Models

The linear regression model achieved MAP = 0.006186, which was 28% lower than the best individual model (TF-IDF: 0.008606). Several factors explain this:

1. Extremely Imbalanced Data

- Only 114 relevant documents out of 22,500 pairs (0.5%)
- Linear regression struggles with such sparse positive examples
- The model learns to predict near-zero scores for most documents

2. Small Dataset

- Cranfield has only 1,400 documents

- Relevance judgments are sparse (~10 relevant docs per query on average)
- Insufficient training data for ML to learn meaningful patterns

3. Feature Correlation

- TF-IDF and BM25 scores are highly correlated (both use TF and IDF)
- Correlated features reduce the effectiveness of linear regression
- The negative coefficient for ES_Builtin suggests overfitting

4. Linear Assumption Limitation

- Linear regression assumes a linear relationship between features and relevance
- Relevance may have a non-linear relationship with retrieval scores
- A non-linear model (e.g., SVM or neural network) might perform better

5.2 Why TF-IDF Performed Best

TF-IDF achieved the highest MAP (0.008606) because:

- IDF component gives higher weight to rare, discriminative terms
- For specialized aerospace vocabulary, rare terms are highly informative
- No aggressive smoothing (unlike Laplace LM)

5.3 Why Laplace Performed Worst

Laplace LM had the lowest MAP (0.005515) because:

- Add-1 smoothing adds 1 to every term frequency
- With vocabulary size ~7,667, this creates a flat probability distribution
- Irrelevant terms get artificially high probability

5.4 Negative Coefficient for ES_Builtin

The learned coefficient for ES_Builtin was negative (-0.000738), meaning higher ES scores lowered predicted relevance. This suggests:

- ES_Builtin scores are on a different scale than other models
- The model may have overfit to noise in the training data

- ES_Builtin correlates with other features but adds no unique information

6. Comparison of All Models

Model	MAP	Rank
TF-IDF	0.008606	1
Okapi TF	0.008157	2
BM25	0.007992	3
ES Built-in	0.007986	4
Jelinek-Mercer	0.006262	5
ML Linear	0.006186	6
Laplace	0.005515	7

Key Observations:

1. **Vector space models (TF-IDF, Okapi, BM25)** outperformed language models
2. **BM25 and ES_Builtin** (both BM25 variants) performed similarly (0.00799 vs 0.00799)
3. **Language models with smoothing** (Laplace, Jelinek-Mercer) performed worst
4. **ML linear regression** did not improve over individual models due to data sparsity

7. Conclusion

This assignment successfully implemented a learning-to-rank pipeline using linear regression with features from six retrieval models. While the ML model did not outperform individual models (MAP = 0.006186 vs TF-IDF = 0.008606), the experiment provided valuable insights:

1. **TF-IDF is the most effective model** for the Cranfield dataset (MAP = 0.008606)
2. **Data sparsity is a major challenge** - Only 114 positive examples out of 22,500 pairs
3. **Feature correlation** reduces the effectiveness of linear models
4. **Laplace LM performed worst** due to aggressive add-1 smoothing